

Unit 2 - Linear Algebra Transformations

MSDS 520

Meharry Medical College

Outline

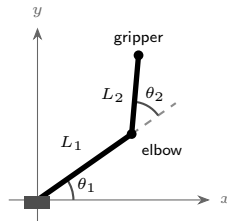
- 1 Matrices and Transformations
- 2 A Matrix as a Function
- 3 What the Columns Tell You
- 4 Types of Transformations
- 5 Length-Preserving Transformations
- 6 Composition and Order
- 7 Beyond Linear: Matrices in Computer Graphics
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Matrices and Transformations

A Problem to Motivate Us

Where is the gripper?

*“A robot arm in a hospital lab has two links: an upper arm of length L_1 from its base to the elbow, and a forearm of length L_2 from the elbow to the gripper. The base turns by θ_1 , and the elbow turns by θ_2 more. For **any** angles, where is the gripper?”*



What the answer will need

- Turning about the **base** is a matrix: vector in, new vector out. This deck builds it.
- The elbow turns about the **elbow**, a point that itself moves. That is **not** a turn about the origin, so no 2×2 matrix does it.
- The last section adds one coordinate, and then the whole arm is **one product of matrices**.

Where This Shows Up in Data Science

Moving points is everywhere

- **Image augmentation.** Classifiers train on rotated, flipped, and stretched copies of every picture: one matrix applied to every pixel's coordinates.
- **Medical imaging.** Lining up two scans of the same patient (**registration**) means finding the turn and stretch that carries one onto the other.
- **Robotics and graphics.** Every robot joint and every turn of a game camera is a rotation matrix, and a chain of joints is a chain of products.
- **Changing coordinates.** Rewriting data on a better set of axes is the same product. **PCA**, this unit's capstone, does exactly that.
- **Neural networks.** Each layer applies a matrix transformation, then a simple nonlinear bend.

Unit 2-3 built the product $A\bar{x}$ as arithmetic. Today it gets a picture: what it does to **space**.

What We Will Build

The plan

- ① **A matrix as a function:** a vector in, a new vector out, and why it is called **linear**.
- ② **What the columns tell you:** where $\hat{i}$ and $\hat{j}$ land, and how to build the matrix you want.
- ③ **Types of transformations:** scaling, shear, rotation (2D and 3D), reflection, permutation, and projection.
- ④ **Length-preserving** transformations: the **orthogonal** matrices.
- ⑤ **Composition and order:** two moves as one matrix, powers, and a look ahead to Units 2-5 and 2-6.
- ⑥ **Beyond linear:** what no matrix can do, and the extra coordinate that lets matrices slide things. It **solves the robot arm**.

Notebooks for this lecture: U2-4_Transformations-1_MatrixAsMap,
U2-4_Transformations-2_NamedTransforms,
U2-4_Transformations-3_UnitCircleAnimation.

A Matrix as a Function

A Matrix Is a Function

Vector in, new vector out

$$\bar{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \xrightarrow{\text{in}} A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \xrightarrow{\text{out}} A\bar{x} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

Same matrix, every input

in $\bar{x}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$
out $A\bar{x}$	$\begin{bmatrix} 2 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 3 \\ 3 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 3 \end{bmatrix}$

Just like a Unit 1 function

- $f(x) = 2x$: a **number** in, a number out. $f(3) = 6$.
- $f(\bar{x}) = A\bar{x}$: a **vector** in, a **new vector** out.
- One fixed rule, the matrix, applied to **every** input.

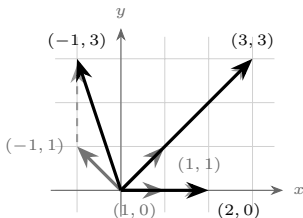
Watching a Matrix Move Vectors

Definition

Multiplying by a matrix A is a function on vectors:

$$f(\bar{x}) = A\bar{x} \quad (\text{input: the vector } \bar{x}; \text{ output: the vector } A\bar{x})$$

A function like this is a **linear transformation**: A does the moving, sending each input vector to a new place.



Gray: inputs. Black: where $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ sends them.

Notebook U2-4_Transformations-1_MatrixAsMap, §1, the point before and after

What goes in, what comes out

- An $m \times n$ matrix takes vectors in $\mathbb{R}^n$ to vectors in $\mathbb{R}^m$.
- 2×2 : the plane to the plane.
Today's case.
- 3×2 : the plane **into** 3D space.

Why “Linear”: the Two Rules and Their Consequences

The transformation respects vector arithmetic

By the distributive and scalar rules for products from Unit 2-3, for every $\bar{u}, \bar{v}$ and every scalar c ,

$$A(\bar{u} + \bar{v}) = A\bar{u} + A\bar{v}, \quad A(c\bar{u}) = cA\bar{u}.$$

Move a sum, or move the parts and add: same answer. That is what **linear** means.

Three consequences, visible in every picture today:

- **The origin stays put:** $A\bar{0} = \bar{0}$.
- **Lines stay lines**, and parallel lines stay parallel.
- **Even spacing stays even:** a square grid becomes a grid of identical parallelograms, never curved or bunched.

Ruled out: a **translation**, sliding the whole plane sideways, moves the origin, so no 2×2 matrix does it. The last section shows what counts, what does not, and the extra-coordinate trick computer graphics uses to bring translations back.

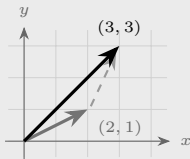
Notebook U2-4_Transformations-3_UnitCircleAnimation, §3, four deformed grids

One Product, Two Interpretations

The basis $\bar{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\bar{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ from Unit 2-2, as the columns of B :

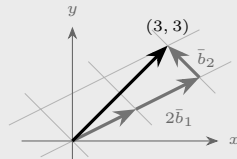
$$B \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

1. A **new vector**, same space



- Input and output are both in **standard** coordinates.
- B **moves** the point $(2, 1)$ to $(3, 3)$: two different vectors.

2. The **same vector**, new basis



- $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is the vector's address in the $\bar{b}$ basis: $2\bar{b}_1 + 1\bar{b}_2$.
- B **translates** that address into standard coordinates. **Nothing moved.**

PCA uses the second: it finds a better basis and rewrites every data point in it.

What the Columns Tell You

Where Do $\hat{i}$ and $\hat{j}$ Land?

Feed the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ the two unit vectors and use the column view:

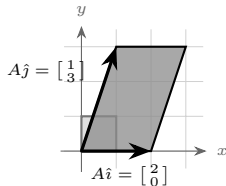
$$A\hat{i} = 1 \begin{bmatrix} a \\ c \end{bmatrix} + 0 \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix}, \quad A\hat{j} = 0 \begin{bmatrix} a \\ c \end{bmatrix} + 1 \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix}.$$

The key fact of this lecture

The columns of A are the places $\hat{i}$ and $\hat{j}$ are sent. Column 1 is the image of $\hat{i}$; column 2 is the image of $\hat{j}$.

Why that tells you everything: every vector is a combination of $\hat{i}$ and $\hat{j}$, and A respects combinations, so

$A(x\hat{i} + y\hat{j}) = x(A\hat{i}) + y(A\hat{j})$. Know where the two unit arrows go and you know where **everything** goes: x of the first column plus y of the second.



$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$: unit square to parallelogram.

Notebook U2-4_Transformations-1_MatrixAsMap, §2, where the unit vectors land

Going Backwards: Build the Matrix You Want

Designing a transformation is two decisions: where should $\hat{i}$ go, and where should $\hat{j}$ go? Write the answers as columns and you are finished.

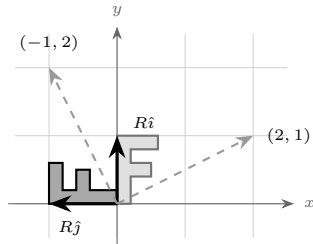
Worked: a quarter turn counterclockwise. Turning $\hat{i}$ a quarter turn points it straight up; turning $\hat{j}$ points it left:

$$\hat{i} \mapsto \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \hat{j} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix} \Rightarrow R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Check on a point: $R \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, which is $(2, 1)$ turned a quarter turn, as the picture confirms.

No formula was looked up; every transformation in the next section was built this way.

Notebook U2-4_Transformations-1_MatrixAsMap, §4, a quarter turn built on purpose



The F before (light) and after R (dark); the columns as arrows.

Exercise: Read It, Then Build It

Try it

- ① For each matrix, say where $\hat{i}$ and $\hat{j}$ land and describe the transformation in words:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}.$$

- ② Write the matrix that sends $\hat{i}$ to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\hat{j}$ to $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Where does it send $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$? And $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$?
- ③ A matrix sends every point to a point on the x -axis. What must its second row be?

Exercise: Solution

Answers

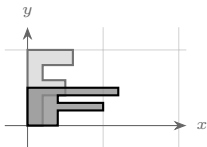
- ① **First:** $\hat{i}$ stays at $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\hat{j}$ goes to $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$: the plane is **flipped across the x -axis**. **Second:** $\hat{i} \mapsto \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\hat{j} \mapsto \begin{bmatrix} 1 \\ 0 \end{bmatrix}$: the two coordinates are **swapped**, a flip across the line $y = x$. **Third:** both arrows are tripled in place: everything is **scaled by 3**, directions unchanged.
- ② Stack the images as columns: $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$. It sends $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to its first column, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ to $2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$. (This matrix turns by 45° and stretches by $\sqrt{2}$: both columns have length $\sqrt{2}$ and sit 45° from where they started.)
- ③ The second entry of every output is $cx + dy$, which must be 0 for all x, y . So $c = d = 0$: the **second row is all zeros**. Both columns then lie on the x -axis, which is where every combination of them must land.

Types of Transformations

Scaling and Shear

Scaling: a diagonal matrix

A **diagonal** matrix (nonzero only on the diagonal) stretches each axis by its own factor: $\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ doubles $\hat{i}$ and halves $\hat{j}$. Equal factors, kI , enlarge uniformly; a negative factor flips that axis.



Light F: the original. Dark F: its image. Check each matrix's columns against where the F's bottom edge and left edge went.

Notebook U2-4_Transformations-2_NamedTransforms, §1, the full panel of named transforms

Shear: slide the top sideways

$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$: $\hat{i}$ stays put, $\hat{j}$ tips over to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Points slide horizontally by an amount proportional to their height, like a deck of cards pushed over.



Rotation by Any Angle

Turn $\hat{i}$ counterclockwise by θ . It is still a unit vector, now at angle θ , so by the definition of sine and cosine it lands at $\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$. The vector $\hat{j}$ stays a quarter turn ahead of it, landing at $\begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$.

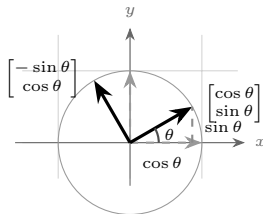
The rotation matrix

Stack the two images as columns:

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

At $\theta = 90^\circ$ this is $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, the quarter turn built by hand.

First piece of the robot arm: turning the whole arm about its base by θ_1 is $R(\theta_1)$. The elbow needs one more idea (last section).



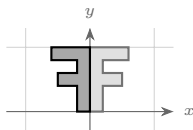
Both unit vectors turned by θ ; their new coordinates are the columns.

Reflection

A reflection flips the plane across a **mirror line** through the origin. Build each one from where $\hat{i}$ and $\hat{j}$ land:

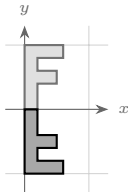
Across the y -axis

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}: \hat{i} \text{ reverses, } \hat{j} \text{ stays.}$$



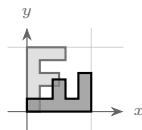
Across the x -axis

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}: \hat{i} \text{ stays, } \hat{j} \text{ reverses.}$$



Across the line $y = x$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}: \hat{i} \text{ and } \hat{j} \text{ trade places.}$$



- Flip twice and you are back: every reflection **undoes itself**.
- The last one **swaps the coordinates**. In 3D, swaps become **permutations**.

Rotations in 3D

In 3D a rotation turns about an **axis**. About each coordinate axis, that axis stays put and the other two turn exactly as in 2D ($c = \cos \theta$, $s = \sin \theta$):

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix}, \quad R_y = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix}, \quad R_z = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Same column rule

- R_z 's last column is $\hat{k}$: the z -axis stays put.
- Its top-left corner is $R(\theta)$ from the plane.
- $R_z(90^\circ)$: $\hat{i} \mapsto \hat{j}$, $\hat{j} \mapsto -\hat{i}$.

Where you meet them

- **Yaw, pitch, roll**: aircraft, drones, phone sensors, game cameras.
- **Order matters**: tip a book then turn it, or turn then tip: two different poses.

The sign flip in R_y is not a typo: it keeps every turn counterclockwise when you look down the axis toward the origin.

Permutations: Reordering Coordinates

A **permutation matrix** is I with its rows reordered: it reorders a vector's entries. The plane has just the swap; 3D has $3! = 6$ orderings.

A cycle is a rotation

$$P \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ c \\ a \end{bmatrix}$$

Three times brings every entry home: $P^3 = I$, a turn by 120° about the line through $(1, 1, 1)$.

A swap is a reflection

$$S \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ a \\ c \end{bmatrix}$$

Twice brings it back: $S^2 = I$, the mirror across the plane $x = y$.

Already used: the row swap in Gaussian elimination (Unit 2-1)

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \left[\begin{array}{cc|c} 0 & 2 & 4 \\ 1 & 1 & 3 \end{array} \right] = \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 2 & 4 \end{array} \right]$$

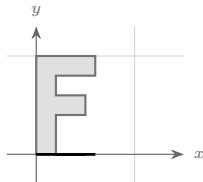
$R_1 \leftrightarrow R_2$, done to get a zero out of the pivot spot, **is** a permutation matrix on the left. Shuffling a data table's rows is the same idea.

Projection: Flattening onto a Line

Projection onto the x -axis

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$$

It keeps $\hat{i}$ and crushes $\hat{j}$ to $\bar{0}$: every point drops straight down onto the x -axis.



Projection destroys information

- The whole F lands on a segment: many inputs share each output.
- So from the segment there is **no way to recover** the F, and the move cannot be undone. The identity, by contrast, moves nothing at all.
- Unit 2-5 measures that collapse with the **determinant** and asks when a transformation can be reversed with an **inverse**.

Exercise: Types of Transformations

Try it

- ① Rotate the point $(2, 0)$ by 90° counterclockwise using $R(90^\circ)$. Then rotate $(1, 1)$ by 45° using $R(45^\circ)$ (take $\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}$). Sketch both.
- ② Apply the shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ to the four corners of the unit square, $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$, and describe the shape you get.
- ③ Write the matrix that reflects across the line $y = -x$, and check it on $(3, 5)$.
- ④ Where does the projection $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ send $(3, 5)$, $(3, -2)$, and $(3, 0)$?
What does that tell you about undoing it?
- ⑤ Apply the cycle $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ to $(1, 2, 3)$, then to the result, then once more. What happens?

Exercise: Solution

Answers

- ① $R(90^\circ) \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$: from the x -axis straight up to the y -axis.
 $R(45^\circ) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1-1 \\ 1+1 \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix}$: the diagonal point turns onto the y -axis, keeping its length $\sqrt{2}$.
- ② $(0, 0) \mapsto (0, 0)$, $(1, 0) \mapsto (1, 0)$, $(1, 1) \mapsto (2, 1)$, $(0, 1) \mapsto (1, 1)$. The bottom edge stays, the top edge slides one unit right: a **parallelogram** leaning right, with the same base and height as the square.
- ③ The mirror $y = -x$ sends $\hat{i}$ to $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$ and $\hat{j}$ to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, so the matrix is $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$. It sends $(3, 5)$ to $(-5, -3)$: swap the coordinates and negate both.

Exercise: Solution (continued)

Answers

- ④ All three go to $(3, 0)$. Three inputs, one output, so **no function can send $(3, 0)$ back** to the right one: the projection cannot be undone. That is what “destroys information” means, and a zero determinant will detect it in Unit 2-5.
- ⑤ Applying P three times: $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Each step moves every entry up one place, the top one wrapping to the bottom. After three steps it is **back where it started**: $P^3 = I$, three turns of 120° making a full turn.

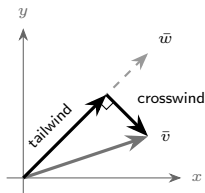
Projection onto Any Line: the Outer Product

So far we have only projected onto the x -axis. Unit 2-2 projected onto a **slanted** line.

Recall from Unit 2-2: wind $\bar{v}$, heading $\bar{w}$

$$\bar{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \hat{w} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

- **Tailwind:** $(\bar{v} \cdot \hat{w}) \hat{w} = 2\sqrt{2} \hat{w} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$
- **Crosswind:** $\bar{v} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$



Regroup the same arithmetic as a matrix

The dot product is a row times a column, $\bar{v} \cdot \hat{w} = \hat{w}^\top \bar{v}$, a 1×1 number, so it can sit on either side of $\hat{w}$. Then regroup (products are associative):

$$(\bar{v} \cdot \hat{w}) \hat{w} = \hat{w} (\hat{w}^\top \bar{v}) = (\hat{w} \hat{w}^\top) \bar{v} = P \bar{v}.$$

Row times column is a number (**inner** product); **column times row** is a 2×2 **matrix**, the **outer product**.

The Tailwind Matrix and the Crosswind Matrix

Tailwind: $P = \hat{w} \hat{w}^\top$, the projection onto the line through $\bar{w}$

Entry (i, j) of the outer product is $\hat{w}_i \hat{w}_j$, and here every one is $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$:

$$P = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$P\bar{v} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3/2 + 1/2 \\ 3/2 + 1/2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Crosswind: $I - P$, because $\bar{v} - P\bar{v} = I\bar{v} - P\bar{v} = (I - P)\bar{v}$

$$I - P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

$$(I - P)\bar{v} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3/2 - 1/2 \\ -3/2 + 1/2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Same tailwind $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and same crosswind $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ as Unit 2-2. And $P + (I - P) = I$: every vector splits into tailwind plus crosswind.

Two Ways to Project

	Dot-product way (Unit 2-2)	Matrix way (today)
Tailwind	$(\bar{v} \cdot \hat{w}) \hat{w}$	$P\bar{v}, \quad P = \hat{w} \hat{w}^\top$
Crosswind	$\bar{v} - (\bar{v} \cdot \hat{w}) \hat{w}$	$(I - P) \bar{v}$
The work	redone for each vector	build P once , apply to all

What the matrix way buys

- **A transformation** like the others in this section: one matrix moves every point at once.
- **Projection onto the x -axis is the special case** $\hat{i}\hat{i}^\top = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$: the shadow on the x -axis.
- $I - P$ is a **projection too**, onto the perpendicular line through $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. P forgets the crosswind; $I - P$ forgets the tailwind.
- **Read the columns**: both are $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$, so $\hat{i}$ and $\hat{j}$ both land on the line through $\bar{w}$.

Length-Preserving Transformations

Which Transformations Keep Lengths?

The question: when is $\|A\bar{x}\| = \|\bar{x}\|$ for every $\bar{x}$? (A rotation: yes. A shear: no.)

The test: $A^\top A = I$

$$\|A\bar{x}\|^2 = (A\bar{x})^\top (A\bar{x}) = \bar{x}^\top \underbrace{A^\top A}_{=I} \bar{x} = \bar{x}^\top \bar{x} = \|\bar{x}\|^2$$

A matrix with $A^\top A = I$ is **orthogonal**, and it keeps every length.

What it says about the columns

Entry (i, j) of $A^\top A$ is column i dotted with column j . So:

- every column has **length** 1;
- the columns are **perpendicular**.

An **orthonormal basis** (Unit 2-2).

Check: $R(\theta)$

- Lengths: $\cos^2 \theta + \sin^2 \theta = 1$.
- Dot product:
 $-\cos \theta \sin \theta + \sin \theta \cos \theta = 0$.

Every rotation is orthogonal.

Notebook U2-4_Transformations-2_NamedTransforms, §2, lengths measured after each matrix

Sorting the Types

Orthogonal (length-preserving)

Rotations, reflections, and permutations. Each column is a unit vector and the two are perpendicular, so $A^T A = I$. Angles are preserved too, since $\bar{u}^T A^T A \bar{v} = \bar{u}^T \bar{v}$. Shapes come out **congruent**: turned or flipped, never distorted.

Not orthogonal

- **Scaling** $\begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix}$: $A^T A = \begin{bmatrix} 4 & 0 \\ 0 & 1/4 \end{bmatrix} \neq I$; lengths along x double.
- **Shear** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$: the second column has length $\sqrt{2}$; the top of the F got longer.
- **Projection** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$: the second column is $\bar{0}$; some vectors lose their length entirely.

Why data science cares: PCA's change of basis is orthogonal, so distances between data points are exactly what they were. The new coordinates are a rotation of the old, not a distortion.

Exercise: Orthogonal or Not?

Try it

For each matrix, compute $A^T A$ and say whether the transformation preserves length. For the ones that do, say what they do geometrically.

$$A_1 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}.$$

Then: without any computation, is $2A_1$ orthogonal?

Exercise: Solution

Answers

- ① $A_1^\top A_1 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. **Orthogonal**: it is the quarter-turn rotation $R(90^\circ)$.
- ② $A_2^\top A_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \neq I$. **Not orthogonal**: the shear stretches its second column to length $\sqrt{2}$.
- ③ Column lengths: $\sqrt{\frac{9}{25} + \frac{16}{25}} = 1$ for both; dot product $-\frac{12}{25} + \frac{12}{25} = 0$. So $A_3^\top A_3 = I$: **orthogonal**. It is $R(\theta)$ with $\cos \theta = \frac{3}{5}$, $\sin \theta = \frac{4}{5}$, a rotation by about 53° . (The 3-4-5 triangle again.)
- ④ **No**. $2A_1$ sends $\hat{i}$ to a vector of length 2. Algebraically $(2A_1)^\top (2A_1) = 4A_1^\top A_1 = 4I \neq I$. Scaling any orthogonal matrix by $k \neq \pm 1$ breaks the property.

Composition and Order

Two Moves in a Row Is One Matrix

Composition is the product

Apply B first, then A : the point $\bar{x}$ goes to $B\bar{x}$, then to $A(B\bar{x}) = (AB)\bar{x}$. So the two-step move is the **single matrix** AB , with the **first** move on the **right**, next to $\bar{x}$.

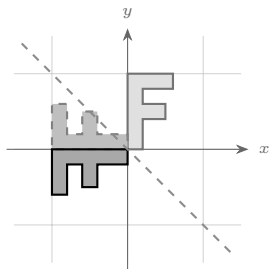
Worked: a quarter turn $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and a flip across the x -axis $F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, in both orders:

$$\text{rotate then flip: } FR = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}, \quad \text{flip then rotate: } RF = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

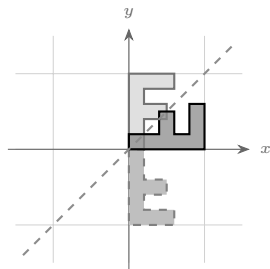
They differ, and the columns say how. RF swaps the coordinates: a reflection across $y = x$. FR sends $\hat{i} \mapsto \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ and $\hat{j} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix}$: a reflection across $y = -x$. That is $AB \neq BA$ from Unit 2-3, now visible as a fact about the world rather than a quirk of arithmetic.

Notebook U2-4.Transformations-2.NamedTransforms, §3, both orders side by side

The Two Orders, Drawn



Rotate, then flip: FR . Light: start. Dashed: after R . Dark: after F . The result is the start reflected across $y = -x$.



Flip, then rotate: RF . Light: start. Dashed: after F . Dark: after R . The result is the start reflected across $y = x$.

Same two moves, opposite order, different final position. Read the dark F's columns off either picture and you recover the product computed on the previous slide.

The Same Move Again and Again: Powers

Applying one transformation n times is the matrix power A^n from Unit 2-3, and knowing the type of transformation makes the powers easy to predict.

Three behaviours

- **Rotation cycles.** Four quarter turns are a full turn: $R(90^\circ)^4 = I$. In general $R(\theta)^n = R(n\theta)$.
- **Projection is idempotent.** Flattening a flattened shape changes nothing: $P^2 = P$, so $P^n = P$ for every $n \geq 1$. Once the information is gone, it stays gone.
- **Shear and scaling accumulate.** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ leans further each time, and $(kI)^n = k^n I$ grows or shrinks geometrically.

Check one: $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, and $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

A reflection is its own undo, $F^2 = I$, which makes it the first **inverse** you have met: Unit 2-5 asks which transformations have one.

Looking Ahead: the Circle Becomes an Ellipse

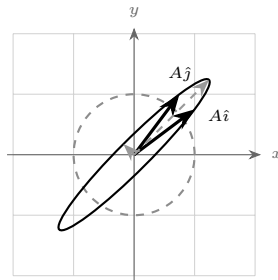
Push every unit vector through a matrix and the unit circle becomes an **ellipse**.

Here $A = \begin{bmatrix} 1 & 0.75 \\ 0.75 & 1 \end{bmatrix}$; its columns sit on the ellipse, being the images of $\hat{i}$ and $\hat{j}$.

Most arrows are **turned** as well as stretched. But the arrows along the ellipse's long and short axes are **not turned at all**: they only change length.

Two questions this picture raises

- By what factor did the **area** grow? (Unit 2-5: the determinant.)
- Which directions are **only stretched**, and by how much? (Unit 2-6: eigenvectors.)



Dashed circle: the unit circle. Solid: its image. Thin arrows: the directions that do not turn.

Notebook U2-4_Transformations-3_UnitCircleAnimation, §2, the swept vector

Exercise: Composition and Powers

Try it

Let $R = R(90^\circ) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

- ① Write the single matrix for “double everything, then turn a quarter turn.” Then the matrix for the opposite order. Are they different? Explain why in this case.
- ② Compute R^2 and say in words what it does. What is R^3 as a rotation?
- ③ Compute PR and RP and describe each geometrically. Which order projects onto the y -axis?

Exercise: Solution

Answers

- ① Double first, then turn: $RS = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. Turn first, then double:

$SR = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. **The same**, because $S = 2I$ and a multiple of the identity commutes with everything: $R(2I) = 2R = (2I)R$. Non-commutativity is the general rule, but uniform scaling is the classic exception.

- ② $R^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$: two quarter turns is a **half turn**, every point sent to its opposite. R^3 is three quarter turns, a rotation by 270° , the same as a quarter turn clockwise.

- ③ $PR = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$: turn first, then flatten onto the x -axis. $RP = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$: flatten onto the x -axis first, then turn that axis up onto the y -axis. So RP is the one that lands everything on the y -axis. Check on $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$: RP gives $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$.

Beyond Linear

Matrices in Computer Graphics

What Counts as Linear, and What Does Not

Linear: some matrix does it

- Scaling and shear.
- Rotation **about the origin**.
- Reflection across a line **through the origin**.
- Projection onto a line **through the origin**.
- Any **chain** of these: a chain is one product.

Not linear: no matrix does it

- **Anything that bends lines:**
 $f(x, y) = (x^2, y)$ turns the line $y = x$ into a curve.
- **Translation:** $f(\bar{x}) = \bar{x} + \bar{t}$ slides everything, so $f(\bar{0}) = \bar{t}$.
- Rotating about any **other** point, or reflecting across a line that **misses** the origin.

The test

A function f is linear when $f(\bar{u} + \bar{v}) = f(\bar{u}) + f(\bar{v})$ and $f(c\bar{u}) = cf(\bar{u})$. Quick first check: is $f(\bar{0}) = \bar{0}$? A translation fails at once, and fails the scaling rule too: $f(2\bar{x}) = 2\bar{x} + \bar{t}$, but $2f(\bar{x}) = 2\bar{x} + 2\bar{t}$.

The Trick: One Extra Coordinate

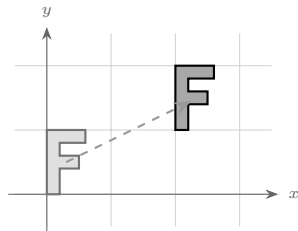
Write (x, y) as $(x, y, 1)$: a slide by $\bar{t} = \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$ becomes a matrix

$$\begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_1 \\ y + t_2 \\ 1 \end{bmatrix} \quad \text{e.g.} \quad \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

Sliding by $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ sends the point $(1, 0)$ to $(3, 1)$.

Why this is allowed

In 3D this matrix is a **shear**: perfectly linear, and it fixes $(0, 0, 0)$. We only look at the slice where the last coordinate is 1, and there the shear looks like a slide. These are **homogeneous coordinates**.



Every point of the F slides by $(2, 1)$.

Every Move Is One 3×3 Matrix

Block form: do A , then slide by $\bar{t}$

$$\begin{bmatrix} A & \bar{t} \\ \bar{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \bar{x} \\ 1 \end{bmatrix} = \begin{bmatrix} A\bar{x} + \bar{t} \\ 1 \end{bmatrix} \quad \text{e.g.} \quad \begin{bmatrix} a & b & t_1 \\ c & d & t_2 \\ 0 & 0 & 1 \end{bmatrix}$$

Multiply it out row by row to check. A plain matrix has $\bar{t} = \bar{0}$; a plain slide has $A = I$.

Rotate about a point $\bar{p}$, not the origin: three moves, one product

$$\underbrace{\begin{bmatrix} I & \bar{p} \\ \bar{0}^\top & 1 \end{bmatrix}}_{\text{3. slide back}} \underbrace{\begin{bmatrix} R & \bar{0} \\ \bar{0}^\top & 1 \end{bmatrix}}_{\text{2. rotate}} \underbrace{\begin{bmatrix} I & -\bar{p} \\ \bar{0}^\top & 1 \end{bmatrix}}_{\text{1. slide } \bar{p} \text{ to the origin}}$$

- Read right to left, as in Composition. A quarter turn about $\bar{p} = (1, 1)$ keeps $(1, 1)$ fixed and sends $(2, 1)$ to $(1, 2)$.
- This is the robot arm's **elbow**: the forearm turns about the elbow, not the base. Next slide.

Solving the Robot Arm

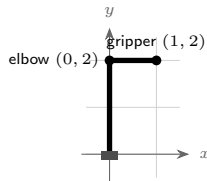
Two kinds of 3×3 matrix

$$\text{Turn}_\theta = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{Slide}_L = \begin{bmatrix} 1 & 0 & L \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The whole arm is one product

$$\text{gripper} = \text{Turn}_{\theta_1} \text{Slide}_{L_1} \text{Turn}_{\theta_2} \text{Slide}_{L_2} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Right to left: lay the forearm flat, bend it by θ_2 , attach it to the upper arm, then swing everything by θ_1 .



Worked: $L_1 = 2$, $L_2 = 1$, $\theta_1 = 90^\circ$, $\theta_2 = -90^\circ$, one matrix at a time

$$(0, 0) \xrightarrow{\text{Slide}_1} (1, 0) \xrightarrow{\text{Turn}_{-90^\circ}} (0, -1) \xrightarrow{\text{Slide}_2} (2, -1) \xrightarrow{\text{Turn}_{90^\circ}} (1, 2)$$

The gripper is at $(1, 2)$, as drawn. Any angles, same four matrices.

Where This Runs Every Day

The extra coordinate is everywhere

- **Video games and 3D graphics.** Points get a fourth coordinate, $(x, y, z, 1)$, and every move is a 4×4 matrix. A graphics card multiplies millions of points by these matrices every frame.
- **Robot arms.** Each joint is one such matrix, and the gripper's position is the product along the chain of joints (**forward kinematics**).
- **Medical imaging.** A common MRI file format (NIfTI) stores a 4×4 matrix that sends each voxel's index to its position in the scanner. Lining up two scans means finding one such matrix.
- **Image augmentation.** Shifting a picture, not just turning it, uses the same trick.

The idea in one line: **add a dimension, and a move that was not linear becomes linear.**

Exercise: One Extra Coordinate

Try it

- ① Write the 3×3 matrix that slides every point by $\begin{bmatrix} -2 \\ 4 \end{bmatrix}$, and apply it to $(1, 1)$.
- ② Where does that slide send $(0, 0)$? Why does this show the slide is not linear **in the plane**?
- ③ Write the 3×3 matrix that **rotates by** 90° about the origin and **then slides** by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Where does $(1, 0)$ go?
- ④ The robot arm with $L_1 = 2$, $L_2 = 1$: where are the elbow and the gripper when $\theta_1 = 0^\circ$ and $\theta_2 = 90^\circ$?

Exercise: Solution

Answers

- ① $\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1-2 \\ 1+4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 5 \\ 1 \end{bmatrix}$: the point $(1, 1)$ lands at $(-1, 5)$.
- ② $(0, 0)$ goes to $(-2, 4)$. A linear transformation always keeps $\bar{0}$ at $\bar{0}$, so the slide is **not linear in the plane**. In 3D, $(0, 0, 0)$ stays put: the 3×3 matrix **is** linear.

Exercise: Solution (continued)

Answers

- ③ Use the block form with $A = R(90^\circ) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $\bar{t} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:
- $$\begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 - 0 + 1 \\ 1 + 0 + 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}. \text{ Check in words: the turn sends } (1, 0) \text{ to } (0, 1), \text{ and the slide moves that to } (1, 1).$$

- ④ $\text{Turn}_{0^\circ} = I$ does nothing, so apply the other three to $(0, 0)$, right to left:

$$(0, 0) \xrightarrow{\text{Slide}_1} (1, 0) \xrightarrow{\text{Turn}_{90^\circ}} (0, 1) \xrightarrow{\text{Slide}_2} (2, 1)$$

The **gripper** is at $(2, 1)$. The **elbow** is the end of the upper arm, $\text{Turn}_{0^\circ}\text{Slide}_2$ applied to $(0, 0)$: $(2, 0)$. The upper arm lies flat along the x -axis and the forearm points straight up.

Review: Transformations

What we covered: a matrix as a map

- ① Multiplying by A is a **function** on vectors, $f(\bar{x}) = A\bar{x}$ (a **linear transformation**): it moves every point of the plane by the same rule.
- ② The rule is **linear**: $A(\bar{u} + \bar{v}) = A\bar{u} + A\bar{v}$ and $A(c\bar{u}) = cA\bar{u}$. So the origin is fixed, lines stay lines, and grids stay evenly spaced. Translation is not linear.
- ③ One product, **two interpretations**: $B\bar{c}$ is a **new vector** in the same space, or the **same vector** with its $\bar{b}$ -basis address $\bar{c}$ rewritten in standard coordinates.
- ④ **The columns of A are where $\hat{i}$ and $\hat{j}$ land.** Read any matrix as a picture; build any transformation by choosing two images and stacking them.
- ⑤ The **types of transformations**: scaling (diagonal), shear, rotation
$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$
 (and R_x, R_y, R_z in 3D), reflection, permutation (a row swap is one), and projection, which destroys information. Onto any line, projection is the **outer product** $P = \hat{w} \hat{w}^\top$, and the leftover is $(I - P) \bar{v}$.

Review: Transformations

What we covered: length, order, and what comes next

- ⑥ A transformation **preserves length** exactly when $A^T A = I$ (**orthogonal**): its columns are an orthonormal basis. Rotations, reflections, and permutations qualify; scaling, shear, and projection do not.
- ⑦ **Two moves in a row are the product** AB , first move on the right. Order matters: rotate-then-flip and flip-then-rotate are different reflections.
- ⑧ **Powers** repeat a move: rotations cycle, a projection is idempotent ($P^2 = P$), shears and scalings accumulate, a reflection undoes itself.
- ⑨ A matrix sends the unit circle to an **ellipse**. How much the area grows is the **determinant** (Unit 2-5); the directions that are only stretched are the **eigenvectors** (Unit 2-6).

Review: Transformations

What we covered: beyond linear

- ⑩ **Linear:** scaling, shear, and rotations, reflections, and projections through the origin, plus any chain of them. **Not linear:** anything that bends lines, and **translations**, which move the origin.
- ⑪ **One extra coordinate:** write (x, y) as $(x, y, 1)$. A slide becomes a 3×3 matrix, every move is $\begin{bmatrix} A & \vec{t} \\ \vec{0}^\top & 1 \end{bmatrix}$, and turning about any point is one product.
- ⑫ **The robot arm:** gripper = $\text{Turn}_{\theta_1} \text{Slide}_{L_1} \text{Turn}_{\theta_2} \text{Slide}_{L_2}$ applied to $(0, 0, 1)$. 3D graphics does the same with 4×4 matrices.

Notebooks: U2-4_Transformations-1, -2, -3. **Previous:** Unit 2-3, Matrices.

Next: Unit 2-5, **Determinants and Inverses.**